| | DBMS Interview Question | One-Line Simple Definition |
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| 1 | **What is DBMS?** | DBMS is software used to create, store, organize, retrieve, and manage data in databases. |
| 2 | **What is a Database?** | A database is an organized collection of related data that can be easily accessed and managed. |
| 3 | **DBMS vs File System?** | DBMS provides structured data management, security, concurrency, and recovery, while a file system mainly stores files without these database features. |
| 4 | **What are the types of DBMS?** | The major types are Hierarchical, Network, Relational, Object-Oriented, and NoSQL DBMS. |
| 5 | **What is RDBMS?** | RDBMS stores data in related tables consisting of rows and columns and follows the relational model. |
| 6 | **DBMS vs RDBMS?** | DBMS manages databases generally, while RDBMS specifically organizes data into related tables and typically enforces relational constraints. |
| 7 | **What is a Table?** | A table is a structured collection of related data organized into rows and columns. |
| 8 | **What is a Record/Tuple?** | A record or tuple represents one complete row of data in a relational table. |
| 9 | **What is a Primary Key?** | A primary key uniquely identifies each record in a table and cannot contain NULL values. |
| 10 | **What is a Foreign Key?** | A foreign key is a column that references a key in another table to establish a relationship between tables. |
| 11 | **What is a Candidate Key?** | A candidate key is a minimal set of attributes that can uniquely identify a record. |
| 12 | **What is a Super Key?** | A super key is any set of attributes that can uniquely identify a record, even if it contains unnecessary attributes. |
| 13 | **What is a Composite Key?** | A composite key is a key made up of two or more columns used together to uniquely identify a record. |
| 14 | **What is an ER Model?** | The Entity-Relationship model represents database entities, their attributes, and relationships between them. |

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| 15 | **What is an Entity?** | An entity is a real-world object or concept about which data is stored, such as Student or Employee. |
| 16 | **What is an Attribute?** | An attribute is a property that describes an entity, such as Student Name, ID, or Age. |
| 17 | **What is Cardinality?** | Cardinality defines how many instances of one entity can be associated with another entity, such as 1:1, 1:N, or M:N. |
| 18 | **What is Normalization?** | Normalization is the process of organizing data to reduce redundancy and prevent data anomalies. |
| 19 | **What is 1NF?** | First Normal Form requires each column to contain atomic values with no repeating groups. |
| 20 | **What is 2NF?** | Second Normal Form requires 1NF and removes partial dependency on part of a composite key. |
| 21 | **What is 3NF?** | Third Normal Form requires 2NF and removes transitive dependency between non-key attributes. |
| 22 | **What is De normalization?** | De normalization intentionally adds some redundancy to improve read performance and reduce complex joins. |
| 23 | **What are ACID Properties?** | ACID means Atomicity, Consistency, Isolation, and Durability, which ensure reliable database transactions. |
| 24 | **What is a Transaction?** | A transaction is a logical unit of database operations that should be completed completely or not at all. |
| 25 | **What is Concurrency Control?** | Concurrency control manages simultaneous transactions to maintain database consistency and correctness. |
| 26 | **What is a Deadlock in DBMS?** | A database deadlock occurs when two or more transactions wait indefinitely for resources locked by each other. |
| 27 | **What is an Index?** | An index is a data structure that improves data retrieval speed but requires additional storage and maintenance. |
| 28 | **Clustered vs Non-Clustered Index?** | A clustered index determines the physical order of table data, while a non-clustered index stores a separate structure pointing to the data. |
| 29 | **What is Database Recovery?** | Database recovery restores the database to a consistent state after failures such as crashes or hardware problems. |

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| 30 | **What is a Database Backup?** | A database backup is a copy of database data used to restore information after data loss or system failure. |

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| | OS Interview Question | One-Line Simple Definition |
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| 1 | **What is an Operating System?** | An Operating System is system software that manages hardware, software resources, and provides services to applications and users. |
| 2 | **What are the main functions of an OS?** | The main functions are process management, memory management, file management, I/O management, security, and resource allocation. |
| 3 | **What is a Kernel?** | The kernel is the core part of the OS that directly manages CPU, memory, devices, and system resources. |
| 4 | **What is a Process?** | A process is a program that is currently executing in memory. |
| 5 | **What is the difference between a Program and a Process?** | A program is passive code stored on disk, while a process is that program actively running in memory. |
| 6 | **What are the states of a Process?** | The main process states are New, Ready, Running, Waiting/Blocked, and Terminated. |
| 7 | **What is PCB?** | Process Control Block (PCB) is a data structure that stores important information about a process, such as its state, PID, and registers. |
| 8 | **What is Context Switching?** | Context switching is the process of saving one process's state and loading another process's state so the CPU can switch between them. |
| 9 | **What is a Thread?** | A thread is the smallest unit of CPU execution within a process. |
| 10 | **Process vs Thread?** | Processes have separate memory spaces, while threads within the same process share memory and resources. |
| 11 | **What is Multithreading?** | Multithreading is the execution of multiple threads within a single process to improve responsiveness and resource utilization. |
| 12 | **What is CPU Scheduling?** | CPU scheduling is the method used by the OS to select which ready process should get the CPU next. |

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|---|---|---|
| 13 | **What are common CPU Scheduling Algorithms?** | Common algorithms are FCFS, SJF, SRTF, Priority, Round Robin, and Multilevel Queue. |
| 14 | **What is FCFS Scheduling?** | First Come First Serve executes processes in the order they arrive in the ready queue. |
| 15 | **What is Round Robin Scheduling?** | Round Robin assigns each process a fixed time slice called a time quantum in a circular manner. |
| 16 | **What is a Race Condition?** | A race condition occurs when multiple threads/processes access shared data concurrently and the final result depends on execution timing. |
| 17 | **What is a Critical Section?** | A critical section is the part of a program where shared resources are accessed and must be protected from concurrent execution. |
| 18 | **What is a Mutex?** | A mutex is a locking mechanism that allows only one thread at a time to access a critical section. |
| 19 | **What is a Semaphore?** | A semaphore is a synchronization mechanism that controls access to shared resources using a counter. |
| 20 | **What is Deadlock?** | Deadlock is a situation where processes wait indefinitely for resources held by each other. |
| 21 | **What are the four necessary conditions for Deadlock?** | The four conditions are Mutual Exclusion, Hold and Wait, No Preemption, and Circular Wait. |
| 22 | **How can Deadlock be handled?** | Deadlock can be handled using prevention, avoidance, detection and recovery, or ignoring it when appropriate. |
| 23 | **What is Memory Management?** | Memory management is the OS function that allocates, tracks, protects, and frees main memory for processes. |
| 24 | **What is Paging?** | Paging divides logical memory into fixed-size pages and physical memory into fixed-size frames for efficient memory allocation. |
| 25 | **What is Segmentation?** | Segmentation divides memory into variable-sized logical segments such as code, data, and stack. |
| 26 | **What is Virtual Memory?** | Virtual memory allows processes to use more memory than physically available by using disk space as an extension of RAM. |

| | OS Interview Question | One-Line Simple Definition |
|---|---|---|
| 27 | **What is Page Fault?** | A page fault occurs when a required page is not present in physical memory and must be loaded from secondary storage. |
| 28 | **What is Thrashing?** | Thrashing occurs when the system spends excessive time swapping pages between RAM and disk instead of executing processes. |
| 29 | **What is a File System?** | A file system is the OS mechanism used to organize, store, retrieve, and manage files and directories on storage devices. |
| 30 | **What is an Interrupt?** | An interrupt is a signal that temporarily stops the CPU's current execution so the OS can handle an important event or request. |

| | CN Interview Question | One-Line Simple Definition |
|---|---|---|
| 1 | **What is a Computer Network?** | A computer network is a group of connected devices that communicate and share data and resources. |
| 2 | **What are the types of networks?** | Common types are PAN, LAN, MAN, and WAN, classified mainly by geographical coverage. |
| 3 | **What is LAN?** | LAN is a network that connects devices within a small area such as a home, office, or building. |
| 4 | **What is WAN?** | WAN is a network that connects devices or networks across large geographical areas, such as countries or continents. |
| 5 | **What is the OSI Model?** | The OSI model is a 7-layer reference model that explains how data communication occurs between networked systems. |
| 6 | **What are the 7 layers of OSI?** | The layers are Physical, Data Link, Network, Transport, Session, Presentation, and Application. |
| 7 | **What is the TCP/IP Model?** | TCP/IP is the practical networking model used on the Internet, commonly represented by Link, Internet, Transport, and Application layers. |
| 8 | **OSI vs TCP/IP?** | OSI is a 7-layer conceptual model, while TCP/IP is the practical protocol architecture used for Internet communication. |

| CN | Interview Question | One-Line Simple Definition |
|---|---|---|
| 9 | **What is a Network Protocol?** | A network protocol is a set of rules that defines how devices communicate and exchange data. |
| 10 | **What is a MAC Address?** | A MAC address is a unique hardware address used to identify a network interface at the Data Link layer. |
| 11 | **What is an IP Address?** | An IP address is a logical address used to identify a device and route packets across a network. |
| 12 | **IPv4 vs IPv6?** | IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses and provides a much larger address space. |
| 13 | **What is a Subnet Mask?** | A subnet mask determines which portion of an IPv4 address represents the network and which represents the host. |
| 14 | **What is a Router?** | A router connects different networks and forwards packets between them based on IP addresses. |
| 15 | **What is a Switch?** | A switch connects devices within a LAN and forwards Ethernet frames using MAC addresses. |
| 16 | **Hub vs Switch?** | A hub broadcasts data to all ports, while a switch forwards data mainly to the intended destination port. |
| 17 | **What is DNS?** | DNS translates human-readable domain names such as google.com into IP addresses. |
| 18 | **What is DHCP?** | DHCP automatically assigns network configuration such as IP address, subnet mask, gateway, and DNS server to devices. |
| 19 | **What is ARP?** | ARP is used in IPv4 networks to discover the MAC address associated with a local IP address. |
| 20 | **What is TCP?** | TCP is a connection-oriented and reliable transport protocol that provides ordered and error-checked data delivery. |
| 21 | **What is UDP?** | UDP is a connectionless transport protocol that provides faster communication without TCP's reliability mechanisms. |
| 22 | **TCP vs UDP?** | TCP provides reliable, ordered delivery, while UDP provides faster communication with no guarantee of delivery or order. |

| | CN Interview Question | One-Line Simple Definition |
|---|---|---|
| 23 | **What is the TCP Three-Way Handshake?** | It establishes a TCP connection using SYN → SYN-ACK → ACK. |
| 24 | **What is a Port Number?** | A port number identifies a specific application or service running on a device. |
| 25 | **What is HTTP?** | HTTP is an application-layer protocol used to transfer web resources between clients and servers. |
| 26 | **HTTP vs HTTPS?** | HTTPS is HTTP protected by TLS encryption, providing confidentiality and authentication for communication. |
| 27 | **What is Routing?** | Routing is the process of determining the path that packets should take from source to destination. |
| 28 | **What is a Firewall?** | A firewall monitors and filters network traffic according to defined security rules. |
| 29 | **What is a Proxy Server?** | A proxy server acts as an intermediary between a client and another server, forwarding requests and responses. |
| 30 | **What happens when you enter a URL in a browser?** | The browser performs DNS resolution, establishes a connection, sends an HTTP/HTTPS request, receives the response, and renders the webpage. |